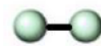
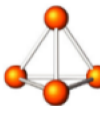
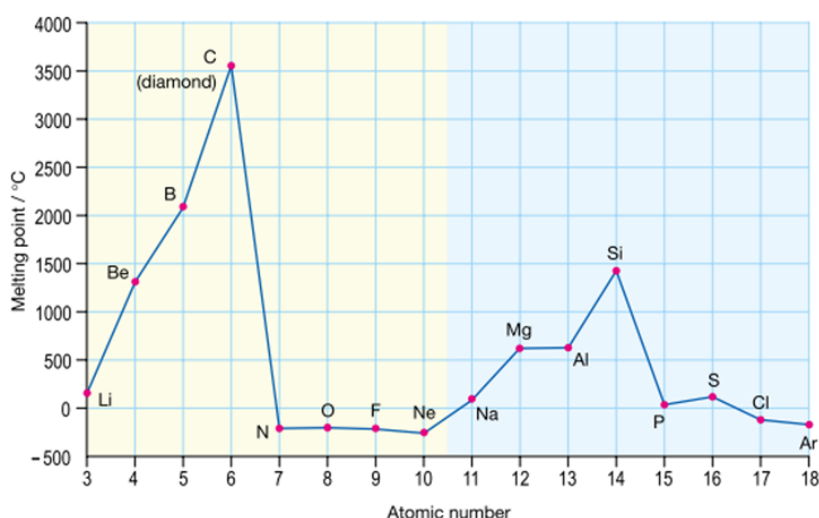


Intensive note (Topic 12: Patterns in chemical worlds)

Structure and bonding of Period 2 elements

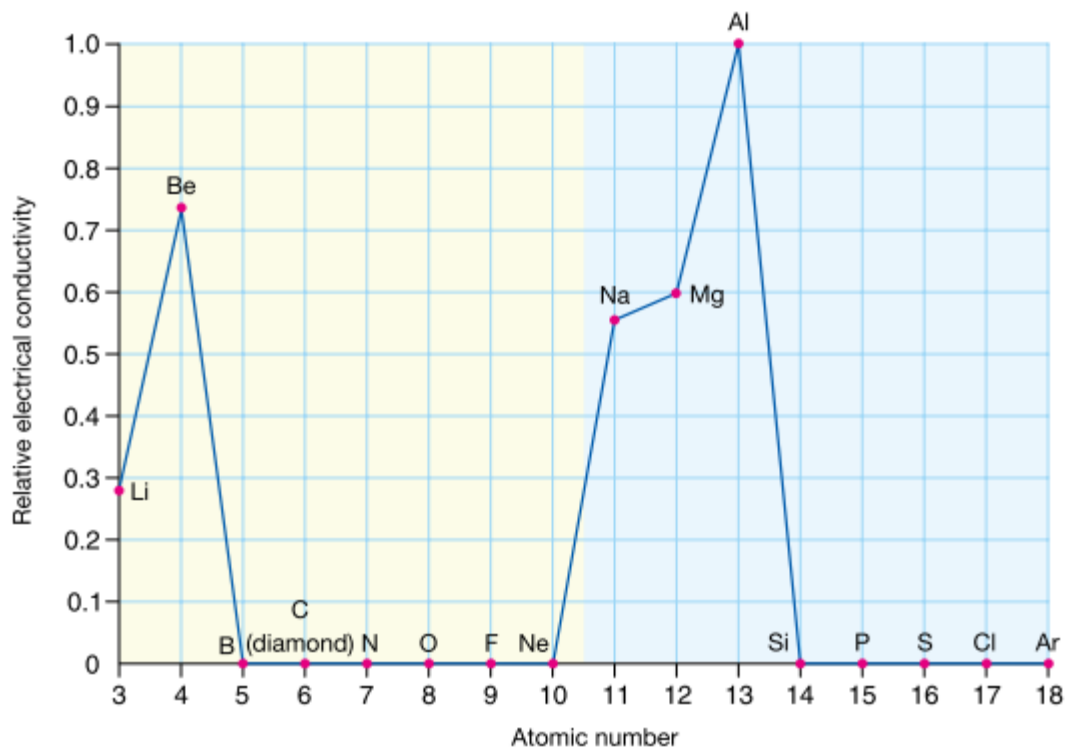
Period 2	I	II	III	IV	V	VI	VII	0
Element	Li	Be	B	C	N	O	F	Ne
Formula	Li	Be	B	C	N ₂	O ₂	F ₂	Ne
Type	Metal		Metalloid	Non-metal				
Structure	Giant metallic structure		Giant covalent structure		Simple molecular			
Bonding	Strong <u>metallic bond</u> between <u>metal ions</u> and <u>delocalized electrons</u>		Strong <u>covalent bond</u> between <u>atoms</u>		1. Strong <u>covalent bond</u> between <u>atoms</u> (except Ne) 2. <u>Weak van der Waals' forces</u> between <u>molecules</u>			
Diagram								
M.p. (°C)	181	1287	2076	3272	- 210	- 219	- 220	- 249

Period 3	I	II	III	IV	V	VI	VII	0
Element	Na	Mg	Al	Si	P	S	Cl	Ar
Formula	Na	Mg	Al	Si	P ₄	S ₈	Cl ₂	Ar
Type	Metal			Metalloid	Non-metal			
Structure	Giant metallic structure			Giant covalent structure	Simple molecular			
Bonding	Strong <u>metallic bond</u> between <u>metal ions</u> and <u>delocalized electrons</u>			Strong <u>covalent bond</u> between <u>atoms</u>	1. Strong <u>covalent bond</u> between <u>atoms</u> (except Ar) 2. <u>Weak van der Waals' forces</u> between <u>molecules</u>			
Diagram								
M.p. (°C)	98	650	660	1414	44	115	- 102	- 189



- Melting point of Be > Li:
The greater the no. of outermost shell electrons per atom, the stronger is the attraction between the metal ions and the delocalized electrons. As Be has more delocalized electrons, metallic bonds in Be are stronger
- Melting point: S > P > Cl > Ar
Molecular size: S₈ > P₄ > Cl₂ > Ar,
Van der Waals' forces between molecules: S₈ > P₄ > Cl₂ > Ar.
- Melting point: N₂ > O₂.
Boiling point: N₂ < O₂

Electrical conductivity of Period 2 and Period 3 elements



- Li, Be, Na, Mg, Al:
They have giant metallic structures. They have delocalized electrons to conduct electricity.
Number of delocalized electrons per atom increases, electrical conductivity increases.
- C, Si:
They have giant covalent structures. Since their atoms are held by strong covalent bonds, electrons are NOT delocalized.
They cannot conduct electricity under room conditions.
[Si can conduct electricity at **elevated temperatures** (as more electrons are delocalized) and the **presence of impurities**]
[**Graphite** (another form of carbon) can conduct electricity as there are **delocalized electrons**]
- N, O, F, Ne, P, S, Cl, Ar:
They have simple molecular structures. They have NO delocalized electrons to conduct electricity.
[F₂ and Cl₂ can conduct electricity in aqueous state as they react with water to give mobile ions]

Intensive note (Topic 12: Patterns in chemical worlds)

Structure and bonding of oxides of Period 3 elements

Period 3	I	II	III	IV	V	VI	VII	0
Formula	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂	Cl ₂ O	---
Structure	Giant ionic			Giant covalent	Simple molecular			---
Bonding	Ionic		Mainly ionic	Covalent				---
Acid-base property	Basic		Amphoteric	Acidic				---
Solubility in water	Soluble	Slightly soluble	Insoluble	Insoluble	Soluble	Soluble	Soluble	---

- Nature of bonding depends on the electronegativity difference
 Electronegativity difference between Na (0.93) and O (3.44) is 2.51. Na₂O therefore is an ionic compound
 Electronegativity difference between Cl (3.16) and O (3.44) is 0.28. Cl₂O therefore is a covalent compound
- In general, oxides of metals are ionic and basic while oxides of non-metals are covalent and acidic

Reactions between oxides of Period 3 elements and water / acid / alkali

	Reaction with water	Reaction with acid (HCl)	Reaction with base (NaOH)
Na ₂ O	Na ₂ O + H ₂ O → 2NaOH 2NaOH ⇌ 2Na ⁺ + 2OH ⁻	Na ₂ O + 2HCl → 2NaCl + H ₂ O Na ₂ O + 2H ⁺ → 2Na ⁺ + H ₂ O	Basic oxide, no reaction
MgO	MgO + H ₂ O → Mg(OH) ₂ Mg(OH) ₂ ⇌ Mg ²⁺ + 2OH ⁻	MgO + 2HCl → MgCl ₂ + H ₂ O MgO + 2H ⁺ → Mg ²⁺ + H ₂ O	Basic oxide, no reaction
*Al ₂ O ₃	Insoluble	Al ₂ O ₃ + 6HCl → 2AlCl ₃ + 3H ₂ O Al ₂ O ₃ + 6H ⁺ → 2Al ³⁺ + 3H ₂ O	Al ₂ O ₃ + 2NaOH + 3H ₂ O → 2NaAl(OH) ₄ Al ₂ O ₃ + 2OH ⁻ + 3H ₂ O → 2Al(OH) ₄ ⁻
SiO ₂	Insoluble	Acidic oxide, no reaction	SiO ₂ + 2NaOH → Na ₂ SiO ₃ + H ₂ O SiO ₂ + 2OH ⁻ → SiO ₃ ²⁻ + H ₂ O
P ₄ O ₁₀	P ₄ O ₁₀ + 6H ₂ O → 4H ₃ PO ₄ H ₃ PO ₄ ⇌ 3H ⁺ + PO ₄ ³⁻	Acidic oxide, no reaction	P ₄ O ₁₀ + 12NaOH → 4Na ₃ PO ₄ + 6H ₂ O P ₄ O ₁₀ + 12OH ⁻ → 4PO ₄ ³⁻ + 6H ₂ O
SO ₂	SO ₂ + H ₂ O → H ₂ SO ₃ H ₂ SO ₃ ⇌ 2H ⁺ + SO ₃ ²⁻	Acidic oxide, no reaction	SO ₂ + 2NaOH → Na ₂ SO ₃ + H ₂ O SO ₂ + 2OH ⁻ → SO ₃ ²⁻ + H ₂ O
Cl ₂ O	Cl ₂ O + H ₂ O → 2HOCl HOCl ⇌ H ⁺ + OCl ⁻	Acidic oxide, no reaction	Cl ₂ O + 2NaOH → 2NaOCl + H ₂ O Cl ₂ O + 2OH ⁻ → 2OCl ⁻ + H ₂ O

*Since Al₂O₃ can react with both acid and base, it is classified as an amphoteric oxide (shows both acidic and basic properties)

Intensive note (Topic 12: Patterns in chemical worlds)

Transition metals

- Transition metals occupy the central block of the Periodic Table between Group II and Group III in Periods 4, 5, 6, 7.
- Properties of transition metals:
 1. They share some common properties of main group metals (e.g. malleable, ductile)
 2. Transition metals are generally stronger, harder, have higher melting points and densities than main group metals
 3. Transition metals are generally less reactive than main group metals

Characteristics of transition metals

1. Formation of colored aqueous ions

Co ²⁺	Cr ₂ O ₇ ²⁻	CrO ₄ ²⁻	Fe ³⁺	Fe ²⁺	Cr ³⁺	Ni ²⁺	Cu ²⁺	MnO ₄ ⁻	Mn ²⁺
Pink	Orange	Yellow	Yellow	Green	Green	Green	Blue	Purple	Pale pink

2. Variable oxidation states in compounds

A variable number of electrons can be removed from the outermost shell and the next inner shell of the atoms of transition metals.

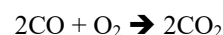
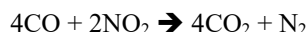
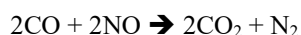
Oxidation states	Transition metals									
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
			+1	+1	+1		+1	+1	+1	
		+2	+2	+2	+2	+2	+2	+2	+2	+2
	+3	+3	+3	+3	+3	+3	+3	+3	+3	
		+4	+4	+4	+4	+4	+4	+4		
			+5	+5	+5	+5	+5			
				+6	+6	+6				
					+7					

Key: the more common oxidation state(s) of the element

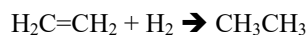
- Metal ions in high oxidation states tend to act as oxidizing agents (undergo reduction)
(Cr₂O₇²⁻, O.N. of Cr = +6)
(MnO₄⁻, O.N. of Mn = +7)
*MnO₄⁻ will be reduced to MnO₂ under alkaline conditions
- Metal ions in low oxidation states tend to act as reducing agents (undergo oxidation)
(Fe²⁺, O.N. of Fe = +2)
(Cu⁺, O.N. of Cu = +1)

3. Catalytic properties of transition metals and their compounds

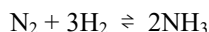
A. Catalytic converter [Platinum (Pt) / Rhodium (Rh)]



B. Catalytic hydrogenation [Nickel(Ni) / Palladium (Pd) / Platinum (Pt)]



C. Haber process [Iron(Fe)]



D. Contact process [Vanadium(V) oxide (V₂O₅)]



Transition metals in our body

1. Cobalt (synthesis of vitamin B₁₂)
2. Manganese (Bones and cartilages)
3. Zinc (Growth and reproductive hormones)
4. Iron (Haemoglobin)